### 7.2 Orsat apparatus : percentage of $CO_2$

*Describe the apparatus that is commonly used for the analysis of dry exhaust gases. Discuss the probable sources of error in such an analysis.*

*A boiler burns coal of the following composition : C = 88% $H_2$ = 3·8%, $O_2$ = 2·2% and the remainder ash. On a particular occasion the percentage of $CO_2$ passing up the chimney was 10%. The temperature of chimney gases was then 250°C. If sample of this gas is analysed by the Orsat apparatus at room temperature, what percentage of $CO_2$ would you expect, assuming complete combustion of the fuel.*

**Solution.** For theory— see text.

Total exhaust gases = $CO_2 + H_2O + O_2 + N_2$

$$\text{Actual percentage of } CO_2 = \frac{CO_2 \times 100}{CO_2 + H_2O + O_2 + N_2}$$

$n = m/M$

| *Mass of constituents per kg of fuel* (*a*) | *Mass of products of combustion* (*b*) | | *Molar mass* (*c*) | | *Proportional volume* (*d*) = (*b*)/(*c*) $(n = m/M)$ | |
|---|---|---|---|---|---|---|
| | $CO_2$ | $H_2O$ | $CO_2$ | $H_2O$ | $CO_2$ | $H_2O$ |
| $C = 0{\cdot}880$ | $0\cdot88\times\frac{44}{12} = m$ | | 44 | | $0{\cdot}0733 = \left(\frac{.88}{12}\right)$ | |
| $H_2 = 0{\cdot}038$ | | $0{\cdot}038\times9$ | | 18 | | $0{\cdot}019 = \frac{.038\times9}{18}$ |
| $O_2 = 0{\cdot}022$ | | | | | | |

$$10 = \frac{0\cdot0733\times100}{0\cdot0733+0\cdot019+O_2+N_2}, \quad \therefore\ O_2+N_2 = 0\cdot641$$

Actual

$C + O_2 \rightarrow CO_2$

12 kg 32 kg 44 kg

.88 $\frac{44}{12}\times .88$

In Orsat apparatus only dfg is measured

$$\therefore \quad \text{Percentage of } CO_2 = \frac{CO_2\times100}{CO_2+O_2+N_2} = \frac{0\cdot0733\times100}{0\cdot0733+0\cdot641} = 10\cdot28\% \qquad \textbf{Ans.}$$

$H_2 + \frac{1}{2}O_2 \rightarrow H_2O$

2 kg 16 kg 18

**7.4 Percentage excess air ; $c_p$, given fuel and dfg analysis**

*The coal supplied to a boiler furnace has the following ultimate analysis :*

*Carbon 82 per cent, hydrogen 5·4 pr cent, oxygen 7 per cent, nitrogen 0·5 per cent, sulphur 0·6 per cent, moisture 1 per cent and the rest is ash. The volumetric analysis of the dry flue gases is $CO_2$ 9·6 per cent, CO 1·2 per cent, $N_2$ 81·4 per cent and $O_2$ 7·8 per cent. Determine,*

*(a) percentage of excess air supplied to the boiler furnace (b) mean specific heat of the dry flue gases.*

*Given the specific heats at constant pressure ; $CO_2$ = 0·879, $O_2$ = 0·9 13, $N_2$ = 1·022 and CO = 1·038 in kJ/kg K units.*

**Solution.** (*a*) Mass of air required per kg of fuel for complete combustion

$$= \frac{100}{23}\left[0\cdot82\times\frac{8}{3}+0\cdot054\times8+0\cdot\underset{.006}{06}\times1-0\cdot07\right]=11\cdot11\text{ kg}$$

| *Volume of constituents per mol of dfg* (*a*) | *Molar mass* (*b*) | *Proportional mass of constituents* (*c*) = (*a*) × (*b*) | *Mass per kg of flue gas* (*d*) = (*c*)/Σ(*c*) | *Mass of carbon per kg of flue gas* (*e*) |
|---|---|---|---|---|
| $CO_2$ = 9·6 | 44 | 9·6 × 44 = 422·4 | 0·14160 | $\frac{0\cdot1416\times12}{44}=0\cdot03862$ |
| CO = 1·2 | 28 | 1·2 × 28 = 33·6 | 0·01126 | $\frac{0\cdot01126\times12}{28}=0\cdot00482$ |
| $N_2$ = 81·4 | 28 | 81·4 × 28 = 2279 | 0·76350 | |
| $O_2$ = 7·8 | 32 | 7·8×32 = 249·6 | 0·08364 | |
| Total = 100·0 | | 2984·6 | 1·00000 dfg | 0·04344 |

$$\text{Mass of dry flue gases per kg of fuel}=\frac{\text{mass of carbon per kg of fuel}}{\text{mass of carbon per kg of flue gases}}=\frac{0\cdot82}{0\cdot04344}=18\cdot88\text{ kg}$$

From table above : Mass of CO per kg fuel = 18·88 × 0·01126 = 0·2126 kg

Mass of unused $O_2$ per kg of fuel = $\cancel{12\cdot88}$ 18·88 × 0·08364 = 1·5791 kg

$$\text{Mass of } O_2 \text{ required for burning of CO }=\frac{0\cdot2126\times4}{7}=0\cdot1215\text{ kg}$$

Excess $O_2$ per kg of fuel = 1·5791 – 0·1215 = 1·4576 kg

$$\therefore\ \text{Excess air }=\frac{1\cdot4576\times100}{23}=6\cdot3374\ \text{ or Percentage exces air }=\frac{6\cdot3374}{11\cdot11}\times100=57\cdot05\%\quad \textbf{Ans.}$$

(*b*) Mean specific heat of dry flue gases

| *Constituents* (*a*) | *Mass per kg of dry flue gas* (*b*) | *Specific heat of constituents at constant pressure* (*c*) | *Heat per kg of dry flue gas* (*d*) = (*b*) × (*c*) |
|---|---|---|---|
| $CO_2$ | 0·14160 | 0·879 | 0·1245 |
| CO | 0·01126 | 0·913 | 0·0103 |
| $N_2$ | 0·76350 | 1·022 | 0·7803 |
| $O_2$ | 0·08364 | 1·038 | 0·0868 |
| Total | 1·00000 | | 1·0019 |

Mean value of specific heat of dry flue gas, $c_p$ = 1·0019 kJ/kg K **Ans.**

## 7.9 Stoichiometric Air/fuel ratio ; mass and volumetric analysis of products

*A fuel having formula $C_8H_{18}$ is burnt with chemically correct amount of air at atmospheric pressure. Write the reaction equation and find the stoichiometric air fuel ratio. Also calculate the percentage by mass and volume of products at atmospheric pressure.*

**Solution.** Since the combustion is theoretically complete with the amount of stoichiometric air the products of combustion will contain only $CO_2$ and $H_2O$ and $N_2$ brought with the air. The combustion equation is written. The amount of $O_2$ required is found by first balancing the C and H atoms on each side of the equation and then summing the number of atoms of O in the products. The amount of $H_2$ is then amount associated with the $O_2$ supplied from the air. $N_2$ does not take part in the combustion, so will remain same, before and after combustion.

Reaction equation is thus

$$C_8H_{18} + 12\tfrac{1}{2}\ O_2 \rightarrow 8CO_2 + 9H_2O$$

$$1\text{ mol}\quad C_8H_{18} + 12\tfrac{1}{2}\text{ mol } O_2 \rightarrow 8\text{ mol } CO_2 + 9\text{ mol } H_2O$$

$$114\text{ kg}\quad C_8H_{18} + 400\text{ kg } O_2 \rightarrow 352\text{ kg } CO_2 + 162\text{ kg } H_2O$$

Since air contains 23 percent of $O_2$ by mass

$$\text{Stoichiometric air/fuel ratio} = \frac{400 \times 100}{114 \times 23} = 15 \cdot 256$$ **Ans.**

Reaction equation considering $N_2$ supplied along with $O_2$ from air is thus

$$C_8H_{18} + 12\tfrac{1}{2}O_2 + \left(\frac{79}{21} \times 12 \cdot 5\right)N_2 \rightarrow 8\,CO_2 + 9H_2O + \left(\frac{79}{21} \times 12 \cdot 5\right)N_2$$

The constituents of products of combustion (in mol) are

$$8\ CO_2 + 9\ H_2O + 47{\cdot}02\ N_2 = 64{\cdot}02\ \text{Mol}$$

We know that the mol fraction of a product equals the volume fraction. Hence volume percentage of constituents in products are :

$$CO_2 = \frac{8}{64{\cdot}02} \times 100 = 12{\cdot}50\%\ ;\ H_2O = \frac{9}{64{\cdot}02} \times 100 = 14{\cdot}06\%\ ;\ N_2 = \frac{47{\cdot}02}{64{\cdot}02} \times 100 = 73{\cdot}44\%$$ **Ans.**

The volumetric composition can be converted to mass composition as shown

| *Volumetric composition in products* (a) | *Molecular mass of constituents* (b) | *Mass composition in products* (c) = (a) × (b) | *Mass percentage* (d) = (c)/Σ(c) × 100 |
|---|---|---|---|
| $CO_2$ = 12·60 | 44 | 550 | 19·24 |
| $H_2O$ = 14·06 | 18 | 253 | 8·85 |
| $N_2$ = 73·44 | 28 | 2056 | 71·91 |
| | | Σ (c) = 2859 | 100·00 |